
DL for ENG
Designed by Remus Teodorescu 2026

LEARNING OBJECTIVES

What You Will Be Able To Do

THE GOALS, IN PLAIN WORDS

- **distinguish** — AI, machine learning and deep learning. Three different things.
- **place** — any model you have built on the physics-to-data spectrum.
- **explain** — why the best fit to the data can be the worst model.
- **state** — what a PINN adds, and what it costs.
- **judge** — when a language model is the right tool, and when it is not.
- **arrive at L4** — able to read a training loop.

WHAT TODAY ASSUMES

from you	curiosity, and an engineering degree
not needed	any prior machine learning
not needed	Python, until L4
do watch	L1 and L2 before the next lecture

NO MATHEMATICS TODAY

One energy function, and one line defining optimisation. The machinery starts in L4 and does not stop until L12.

THE ARC

Where This Sits in Twelve Lectures

TWO PARTS, ONE METHOD

- **L1 – L2** — Python and PyTorch, recorded. Watch before L4.
- **Part 1 · L3 – L6** — what a network is, and how training works. General.
- **Part 2 · L7 – L12** — physics in the loss. Heat, flow, batteries, a car, the grid.
- **the join is L6.1** — a loss term need not contain any data at all

THE TWELVE

L1 – L2	Python and PyTorch — recorded
L3 – L4	what AI is; networks from scratch
L5	convolutional, graph, sequence, attention
L6	training, and what comes after
L7 – L12	physics-informed networks, applied

WHY THE HISTORY IS NOT DECORATION

The field's two collapses were both caused by claiming more than the method could deliver. You are about to be handed a method that is easy to over-claim.

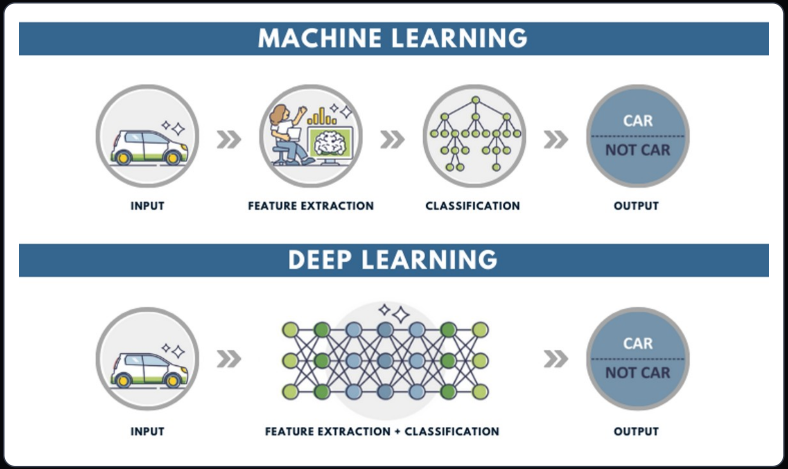
DEFINITIONS

Four Areas, and Why One Ate the Others

THE WORDS, USED PRECISELY ONCE

- **artificial intelligence** — the whole ambition: logic, search, planning. Mostly no learning.
- **machine learning** — decisions from fitted models. Linear regression counts.
- **deep learning** — many layers, fitted by gradient descent. That is all it means.

THE PRACTICAL DIFFERENCE: WHO EXTRACTS THE FEATURES



FOUR AREAS, ONE INSIDE THE NEXT

ARTIFICIAL INTELLIGENCE
search · planning · knowledge representation · reasoning · robotics · language

MACHINE LEARNING
regression · k-means · PCA · SVM · decision trees · random forest · k-NN

NEURAL NETWORKS
perceptron · MLP · Hopfield · Boltzmann machine

DEEP LEARNING
CNN · RNN · GNN · transformers · PINNs — this course

THE ONE TO REMEMBER

A Kalman filter is AI by any 1970s definition, and still the best tool for state estimation in most control rooms. In L12 it is the baseline your neural method must beat — and often will not.

Three Ingredients, Twelve Lectures

EVERYTHING AFTER THIS IS A VARIATION

- **a model** — anything that maps inputs to outputs and can be wrong
- **the three ingredients** — a functional form, a loss, and a way to minimise it
- **the only question** — where the form comes from: physics, or the data

$$\hat{\theta} = \arg\min_{\theta} \mathcal{L}(\theta)$$

Fit the knobs to minimise the wrongness. That is the entire subject.

FUNCTION
flexible,
with knobs

→

LOSS
one number:
how wrong

→

OPTIMISER
turns that into
better knobs

Refer back to these three every lecture. Only the middle one changes in Part 2.

THE SAME THREE, LECTURE BY LECTURE

L4	the function gets deep
L5	the function gets structure
L6.1	the loss gets built properly
L7	the loss gets physics in it
L12	the function is a whole power grid

WHAT ACTUALLY CHANGES IN PART 2

Only the middle one. The function stays a network and the optimiser stays gradient descent. Physics-informed learning is a statement about the loss, and nothing else.

The Neuron, the Winter, and Hopfield

FORTY-THREE YEARS IN ONE SLIDE

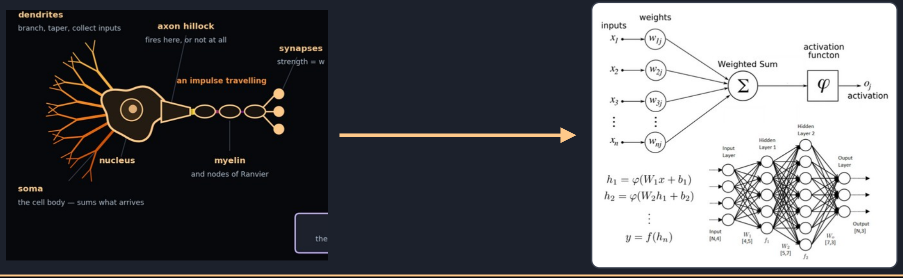
- **1943 · McCulloch and Pitts** — the neuron as a threshold on a weighted sum. A paper in mathematical biology, not computing.
- **1958 · Rosenblatt** — the perceptron, built in hardware at Cornell. It learns to classify from examples.
- **1969 · Minsky and Papert** — one layer cannot do XOR. True of one layer, read as fatal to all. Funding collapses: the first AI winter, twenty years.
- **1982 · Hopfield** — memory as a physical system with an energy; the network settles into its minima. Statistical mechanics.
- **1986 · Rumelhart, Hinton and Williams** — backpropagation trains many layers. Winter ends.

HOPFIELD'S ENERGY

$$E = -\frac{1}{2}\sum_{i,j} w_{ij}\, s_i\, s_j \;-\; \sum_i \theta_i\, s_i$$

Neural networks began in statistical physics: the Hopfield network (1982) and Hinton's Boltzmann machine (1985) are spin-glass energy landscapes. This course takes them back to engineering; the prize that recognised the origin is two slides on.

A REAL NEURON, AND THE UNIT THAT COPIES IT



Why It Worked in 2012 and Not in 1992

THE LONG MIDDLE, AND THEN TWO PAPERS

- **1986 – 2012** — the mathematics barely changed. SVMs won.
- **2012 · AlexNet** — Krizhevsky, Sutskever and Hinton. Not a new idea: 1.2 million labels and two gaming GPUs.
- **the uncomfortable lesson** — a correct method, unusable for twenty-six years
- **2017 · attention** — every language model you have used descends from it

ALEXNET, 2012 · KRIZHEVSKY, SUTSKEVER AND HINTON

IMAGE	CONV 11×11	CONV 5×5	CONV 3×3 ×3	FC	FC	OUT
224×224×3	96 · pool	256 · pool	384 → 256	4096	4096	1000 classes

Eight learned layers, 60 million parameters, trained for six days on two GTX 580 gaming cards. Error on ImageNet: 15.3 %, against 26.2 % for the next entry.

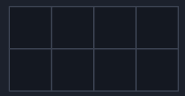
WHAT ARRIVED, AND WHEN

the idea	1986 — and then it waited
the data	ImageNet, 2009, 14 million images
the arithmetic	GPUs, and someone willing to try

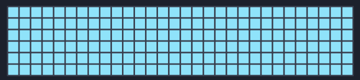
THE ONE TO REMEMBER

Too little data and compute make a correct method look wrong. Part 2 makes the opposite trade: physics in the loss, less data.

GPU VS CPU



CPU · 8 large cores, one instruction each, in sequence



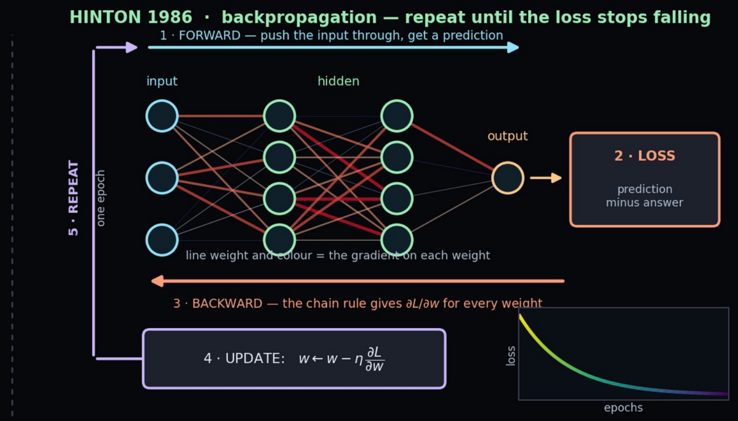
GPU · thousands of small cores, all at once

A matrix product is one operation on thousands of numbers. NVIDIA: founded 1993 by Jensen Huang, Malachowsky and Priem for 3D-game chips; GeForce 256 (1999), the first “GPU”; CUDA (2006), Huang’s bet that those cores could do any arithmetic.

Physics, Awarded to Two Computer Scientists

OCTOBER 2024

- **October 2024** — the Nobel Prize in Physics. Hopfield and Hinton.
- **in physics** — not a category error. Hopfield's model is statistical mechanics.
- **the loan** — the field borrowed its founding ideas from physics
- **Part 2** — is the loan repaid: conservation laws, written into a loss



THE LOOP THAT TRAINS IT

Forward: push the input through, get a prediction. Loss: compare it with the answer. Backward: the chain rule gives the gradient on every weight. Update: nudge each weight against its gradient. Repeat until the loss stops falling. The 1986 idea, and every network in this course is trained this way.

WHAT WAS CITED

Hopfield 1982
associative memory as an energy function

Hinton 1985
the Boltzmann machine, unsupervised learning

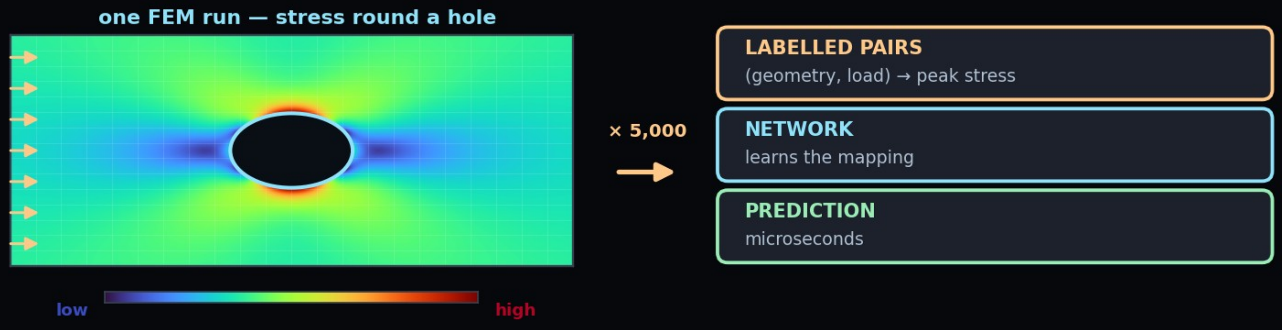
Hinton and Hopfield, Stockholm 2024

"for foundational discoveries and inventions that enable machine learning with artificial neural networks" — the Nobel Prize in Physics 2024, awarded jointly

Supervised Learning

YOU HAVE THE ANSWERS ALREADY

- **supervised** — inputs with known answers. The answers are the supervision.
- **the cost** — someone had to label every example
- **in engineering** — a simulation is a label factory. Run FEM once, learn it forever.
- **Ex03** — is supervised: time in, displacement out



the answers already exist

IN THIS COURSE

Ex_03	fit a curve, three ways
Ex_04	classify, and fail at XOR
Ex_05	images, and a six-bus grid
L11.1	steering angle from a camera

THE VERSION THAT BREAKS THE PATTERN

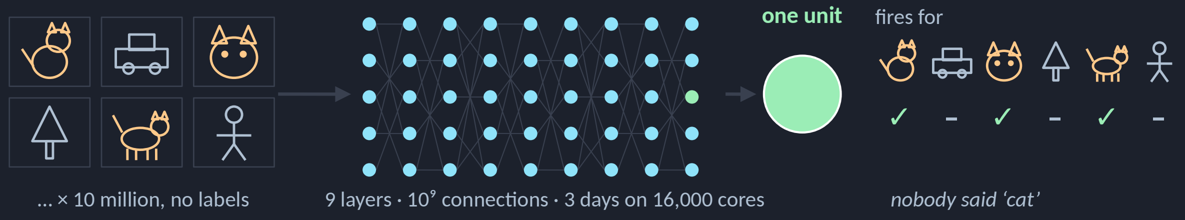
In L7 onward you will train networks with no labelled outputs at all. The supervision comes from a differential equation instead. Everything you learn about overfitting here has to be re-examined there — L6.2 says how.

Unsupervised Learning

YOU HAVE DATA AND NO ANSWERS

- **unsupervised** — no answers. Find the structure already there: clusters, axes, anomalies.
- **the other face** — features learned from unlabelled data. Google, 2012: a network taught itself 'cat' from 10 million unlabelled frames (below)
- **in engineering** — 'normal' from unlabelled logs, then flag departures; features from unlabelled inspection images (blades, PV, thermal), then a few labels
- **the catch** — nothing says the structure found is the one you wanted

GOOGLE BRAIN, 2012 · FRAMES IN, ONE UNIT FIRES FOR CATS



WHAT IS IN SCOPE

PCA	yes — mentioned in L6.2
clustering	yes — briefly
anomaly detection	yes — the engineering case
GANs, diffusion, VAEs	out by decision — reading: UDL ch. 14 to 18

LEARNING FEATURES WITHOUT LABELS

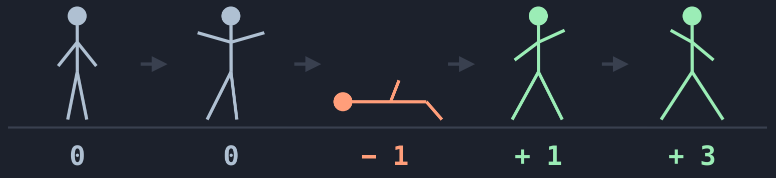
Hinton, Toronto 2006	deep belief nets: layers trained on unlabelled digits, a few labels on top
Google Brain 2012	a 'cat' unit from 10 million unlabelled YouTube frames
language models	pre-trained the same way, on text nobody labelled

Reinforcement Learning

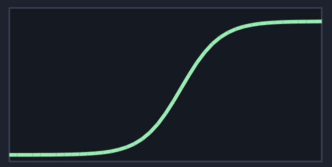
YOU HAVE NEITHER — YOU HAVE CONSEQUENCES

- **reinforcement** — no answers, only consequences. Act, observe, adjust.
- **the difference** — supervised is told the answer, unsupervised is told nothing, reinforcement is told how good its answer was, later: a score, not a correction
- **the reward** — is the whole design problem. Get it wrong and it optimises the wrong thing.
- **the cost** — millions of trials, so it lives in simulators. In engineering: control, scheduling, energy management.

LEARNING TO WALK · TRY, FALL, TRY AGAIN



attempt → consequence → adjust. No labels, no dataset: the floor is the teacher.



reward against attempts: flat, then up, then flat. Millions of them, in a simulator.

THE VOCABULARY, TRANSLATED

state	what the plant is doing
action	the control input
policy	the controller
reward	minus the cost function
episode	one run of the experiment

WHERE YOU WILL MEET IT

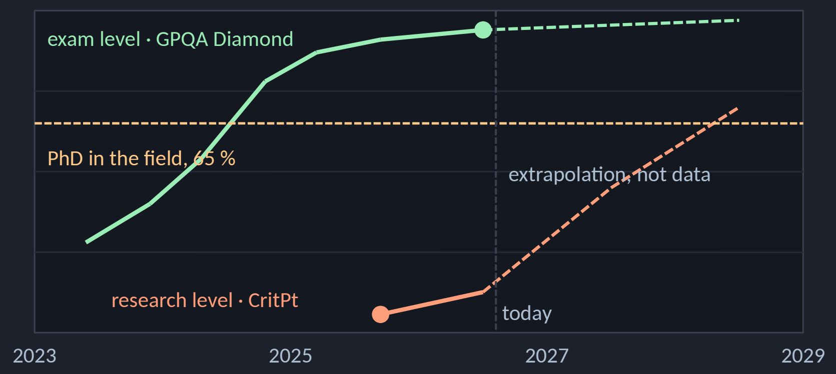
L6.2 teaches the fundamentals and the role of reinforcement in post-training. L11 runs it on a JetRacer, where a bad policy costs you a physical crash rather than a number in a table.

What Language Models Can Do, and What They Cannot

EXAMS PASSED, RESEARCH STARTED, A DIFFERENT CHECK

- **exam level** — GPQA Diamond, 198 Google-proof graduate questions. A PhD scores 65 %; frontier models passed that in late 2024, now low nineties.
- **'research', two meanings** — leaderboards score knowledge and web search: 90+ % in 2026. CritPt scores solving unpublished physics problems: 5.7 % in 2025, 12.6 % with tools. Improving fast.
- **in mathematics they already do research** — Olympiad gold in 2025, open problems since. The check there is a proof, and a machine can verify a proof.
- **in engineering the check is physical** — a measurement, a conservation law, a test rig. No language model can run your experiment.
- **what they produce is text and code** — what you need is a model of your system: numbers with a known error, obeying the physics

EXAMS: PAST THE PhD LINE. RESEARCH: NEXT



THE CONCLUSION

Language models accelerate your research; they do not replace you in it. A language model is fluent about physics; a physics-informed network is constrained by it. The PINN on our oscillator is wrong by 0.59 mm and knows which equation it is solving, because the equation is in its loss. Use the first to build the second.

Where They Help, and the Rule for This Course

USE THEM. THEY ARE VERY GOOD AT SOME THINGS.

- **good** — code, plotting, stack traces, finding the paper you needed
- **good** — algebra, provided you verify it
- **bad** — any number you act on without checking
- **the failure mode** — wrong fluently, at length, without hesitation
- **course rule** — declare use in one line. Defend every physical number yourself.

GOOD USE

code	plotting, refactoring, stack traces
literature	finding and summarising
algebra	derive, then verify

BAD USE

numbers	any figure you have not checked
judgement	which model to defend, and why

THE COURSE RULE

Declare it in one line. Defend every physical number yourself.

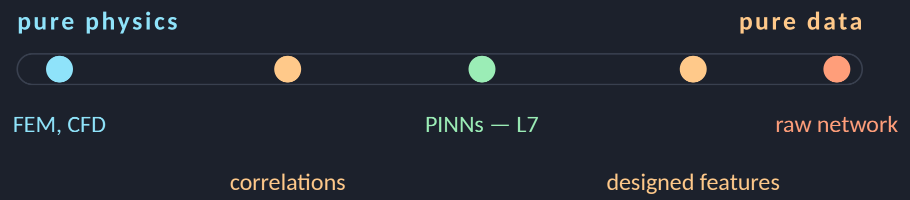
THE CENTRAL IDEA

The Modelling Spectrum

NOT A CHOICE BETWEEN TWO CAMPS

- **not two camps** — a line, and every model sits on it
- **pure physics** — no data, extrapolates, defensible. Slow, and you must know the physics.
- **pure data** — no equations, microseconds. Cannot extrapolate, cannot explain.
- **already hybrid** — a fitted correlation is physics with constants from data

THE LINE, END TO END



pure physics	no data, extrapolates, slow
hybrid	physics as a penalty — L7 to L12
pure data	fast, silent outside its range

THE QUESTION

Never physics or data. How much of each, and which you will defend when the answer surprises someone.

THE CENTRAL IDEA

Where Your Own Models Already Sit

AN EXERCISE FOR YOU

Take a model you have built — for a thesis, a project, a job. Place it on the spectrum, then say what you would need to move it one step towards physics.

FINITE ELEMENT ANALYSIS

Pure physics. Trustworthy, extrapolates, and too slow to put inside an optimisation loop or a controller.

A LOOKUP TABLE FROM TEST DATA

Pure data, and the oldest form of it. Fast and exact where you measured; silent about everywhere else.

A FITTED CORRELATION

Hybrid, and universally accepted. Nusselt as a function of Reynolds and Prandtl is a physical form with three fitted numbers.

AND WHAT THIS COURSE ADDS

A hybrid too — but physics enters as a penalty, not as a form.

You state the equation the answer must satisfy. The optimiser finds a function that does.

Buys: problems with no closed form, and unknown parameters recovered as a by-product.

Costs: you cannot read the physics off the fitted model.

What Deep Learning Is Actually Good At

FOUR THINGS, AND THEY HAVE A COMMON SHAPE

- **interpolation** — superb, where the data is dense. Ex03: 1.8 mm inside the window.
- **high dimensions** — where correlations are real but nobody can write them down
- **speed** — microseconds per evaluation, once trained
- **inverse problems** — recovering a parameter you cannot measure directly

THE COMMON SHAPE

Complicated function, simple objective — that is the shape.

The method supplies the function. You must supply the objective.

Which means the engineering judgement moves entirely into the loss. Choose it badly and you will get a model that optimises exactly what you asked for and nothing you wanted.

In L7 the loss stops being a formality and becomes the design.

When Not To Use It

YOUR DATA IS THIN

Fit a curve instead. A model with a handful of physical parameters is defensible on the data you have; one with thousands is not, however good the training error looks.

YOU NEED A GUARANTEE

Certification wants bounds. A trained network gives an empirical error on the data you happened to test — a different kind of object.

THE PHYSICS IS KNOWN AND CHEAP

If a solver runs in a second, use the solver. A network that took six hours and is wrong unpredictably is not an improvement.

ASK IN THIS ORDER

first	can I write the equation down?
then	is solving it fast enough?
then	do I have data in the hundreds or the millions?
then	who has to believe the answer?
last	and only then, which architecture?

Almost every failed machine-learning project in engineering skipped straight to the last question. The ordering is the content of this slide.

How Much Data — the Engineering Reality

THE NUMBERS YOU READ ABOUT ARE NOT YOUR NUMBERS

- **the honest answer** — it depends how much physics you are willing to write down
- **no physics** — thousands of examples, and it still fails outside them
- **some physics** — Ex03 model 4: 101 noisy points, and 0.59 mm outside the data
- **all physics** — Ex03 model 5: no measurements at all. The equation and x(0) suffice.
- **so 'how much data'** — is the wrong question, asked first

ORDERS OF MAGNITUDE

a thesis test campaign	10^2 runs
a year of plant SCADA	10^6 samples, few events
ImageNet	10^6 labelled images
a modern language model	10^{12} tokens
a PINN in L7	zero labelled outputs

The last row is not a trick. In L7 you will train a network that has never been shown a correct answer, and it will produce one.

THE PROPOSITION

What This Course Does Differently

A NORMAL DEEP LEARNING COURSE

Teaches architectures, then applies them to benchmark datasets with millions of labelled examples.

Measures success by accuracy on a held-out test split.

Treats the loss function as a technical detail chosen from a short list.

Produces engineers who can train a model on data somebody else collected, and who are stuck the moment the data runs out.

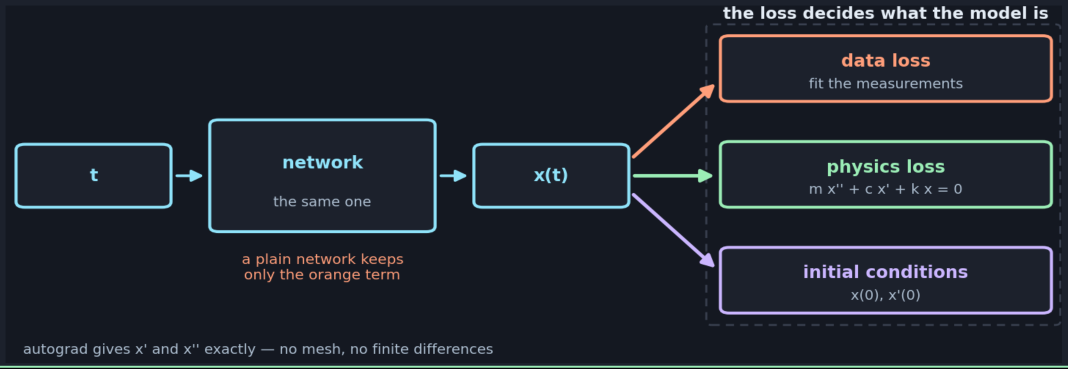
None of this is wrong. It is simply not the situation most engineering problems present.

THIS ONE

Teaches the same architectures — L4 and L5 are not unusual — and then spends six lectures on problems where the data is scarce and the physics is known.

Measures success by whether the answer satisfies a conservation law you can check independently.

Treats the loss as the entire design decision — that is where the physics enters.



The distinctive claim: physics is worth data, and you can put a number on the exchange rate.

Failure Modes — of the Field, Not the Model

BENCHMARK CHASING

A benchmark becomes the goal. The gain is real on that dataset and often vanishes on yours.

IRREPRODUCIBILITY

Results nobody can reproduce, because a seed or an undocumented trick did the work. Common in the young PINN literature. Treat accuracy claims as provisional.

THE CONFIDENT WRONG ANSWER

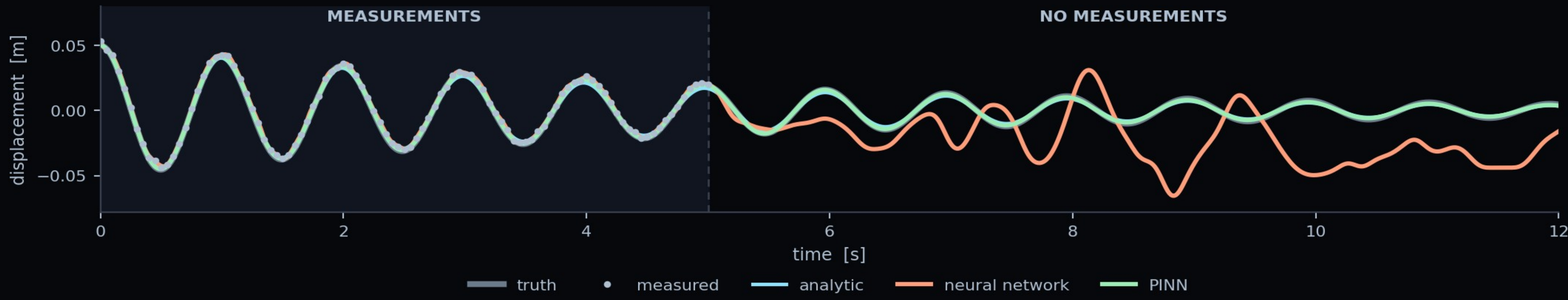
A network always returns something. Nothing tells you the question was outside everything it saw. This is the one that reaches a report.

THE DEFENCES, IN ORDER OF COST

free	fix and report your random seed
cheap	test on a regime you held out
cheap	check a conservation law
real work	compare against a classical baseline
real work	state what you assumed, in writing

Every one of these appears as a marked requirement in an exercise report. They are not good practice added on top — they are how you know the answer is real.

One Vibrating Mass, Four Ways to Predict It



WHAT THE NUMBERS SAY

- **the data** — 5 s from a sensor that is noisy and was never zeroed
- **the network** — agrees with the measurements to 0.62 mm. They are wrong by 1.80 mm.
- **so it fitted the noise** — and is fifty times worse than the PINN outside the window
- **model 5** — no measurements at all. The equation and x(0) are enough.

ERROR OUTSIDE THE DATA

analytic	1.34 mm
regression	3.03 mm
network	30.26 mm
PINN	0.59 mm

THE QUESTION THAT CARRIES THE MARKS

All four fit the data. Which would you put in a design report, and why? The marks are for the reasoning, not the error.

SUMMARY

Key Takeaways

1	**Three ingredients, always**	A parameterised function, a loss, an optimiser. Twelve lectures are variations on these.
2	**The field borrowed from physics**	Hopfield's energy and the Boltzmann machine are statistical mechanics. The 2024 Nobel Prize in Physics says so officially.
3	**It collapsed twice, after over-claiming**	1969 and the late 1990s. The method is genuinely good now; the incentive to over-claim has not changed.
4	**Modelling is a spectrum, not a side**	Physics at one end, data at the other, and almost every real engineering model in between. L7 to L12 live in the middle deliberately.
5	**Knowing when not to use it is the skill**	Forty data points, a required guarantee, or physics that is known and cheap — three good reasons to close the laptop.

Next — L3.2: what the field has actually built, across science and engineering.

OVER TO YOU

Questions

SIX TO TEST WHETHER TODAY LANDED

- **the three words** — which is the subset of which, and where does a Kalman filter sit?
- **your own model** — where on the spectrum, and what would move it towards physics?
- **fit versus truth** — how can the best fit to the data be the worst model?
- **the sensor bias** — which of the five Ex03 models did it damage, and why?
- **model 5** — used no measurements. So what did the measurements contribute?
- **the two AIs** — you must size a heat exchanger by Friday. Which one, and what would you not let it decide?

BEFORE L4

watch	L1 and L2 — L4 assumes them
run	the Ex03 notebook, end to end
bring	the four-model figure to L4

AND ONE WITH NO ANSWER YET

What evidence would convince you a physics-informed model was wrong? If you cannot say, you cannot defend it either.

WHERE THE HOUR LANDED

Two kinds of AI. One writes your code. The other solves your differential equations. This course builds the second.

SOURCES

References and Further Reading

THE TWO BOOKS THIS COURSE USES

Prince (2023–26), Understanding Deep Learning, MIT Press — the main reference throughout. Free PDF at udlbook.github.io/udlbook. Chapter 1 is today's lecture; chapters 2 to 9 are L4 and L6.

Goodfellow, Bengio & Courville (2016), Deep Learning, MIT Press — free at deeplearningbook.org. Used for four things UDL does not contain. Today, chapter 1.2, which is the best short history of the field in print.

Note the trap: Goodfellow chapter 16 is titled “Structured Probabilistic Models Using Graph Theory” and is about graphical models, not graph neural networks. Those are UDL chapter 13, in L5.1.

THE PRIMARY SOURCES BEHIND TODAY

Hopfield (1982), PNAS 79(8), 2554–2558 — four pages, and readable by anyone with an undergraduate physics course.

Rumelhart, Hinton & Williams (1986), Nature 323, 533–536 — backpropagation.

Krizhevsky, Sutskever & Hinton (2012), NeurIPS 25 — AlexNet.

The Nobel Prize in Physics 2024, nobelprize.org — the scientific background document is an excellent 20-page history.

READ IN THIS ORDER

before L3.2	UDL ch. 1 — twenty minutes
if the history caught you	GBC ch. 1.2, then Hopfield (1982)
before L4.1	UDL ch. 2 and 3
for the exercise	the run guide in Ex_03
for interest only	the Nobel background PDF

Both books are free and legally downloadable. Nothing on this slide is behind a paywall, and the two primary papers are open access.

NEXT

L3.2 — What the Field Has Built

Today was what the field is and how it got here. Next time is what it has actually produced — across vision, language, control, and then across science and engineering, which is where this course is going.

BEFORE THE NEXT SESSION

read	UDL chapter 1 — about twenty minutes
run	Ex_03 notebook 00, to check your setup
bring	one model from your own work
think about	where on the spectrum it sits

Bring the model. L3.2 ends by placing the room's own examples on the spectrum, and a lecture theatre full of engineers produces better cases than any set I could invent — but only if you arrive having thought about one.